WORKED SOLUTION

# STEP II 2024, Question 4

Pure • Vectors and Angle Bisectors

**STEP II 2024, Question 4**

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

## The key idea

Scale everything down to unit vectors. If $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ are the unit vectors along $OX$ and $OY$, then $\hat{\mathbf{x}} + \hat{\mathbf{y}}$ is the diagonal of a rhombus, and the diagonal of a rhombus bisects its angle. The given $\mathbf{b}$ is just a positive multiple of $\hat{\mathbf{x}} + \hat{\mathbf{y}}$, so each bisecting vector is a positive multiple of a sum of two unit vectors. That makes the dot products in part (ii) collapse to a single symmetric expression.

## Solution

Throughout, write $\theta$ for the angle $XOY$. Since $O$, $X$, $Y$ are not collinear, $0 < \theta < \pi$, so

$$-1 < \cos\theta < 1.$$

Both strict inequalities get used below.

### Part (i)(a)

Let $\hat{\mathbf{x}} = \mathbf{x}/|\mathbf{x}|$ and $\hat{\mathbf{y}} = \mathbf{y}/|\mathbf{y}|$, so $\hat{\mathbf{x}} \cdot \hat{\mathbf{y}} = \cos\theta$. Then

$$\mathbf{b} = |\mathbf{x}|\mathbf{y} + |\mathbf{y}|\mathbf{x} = |\mathbf{x}||\mathbf{y}|\,(\hat{\mathbf{x}} + \hat{\mathbf{y}}).$$

This is a combination of $\mathbf{x}$ and $\mathbf{y}$, so it lies in the plane $XOY$. It is non-zero, because $\hat{\mathbf{x}} + \hat{\mathbf{y}} = \mathbf{0}$ would mean $\hat{\mathbf{y}} = -\hat{\mathbf{x}}$, which would make $O$, $X$, $Y$ collinear.

Now compare the angles. We have

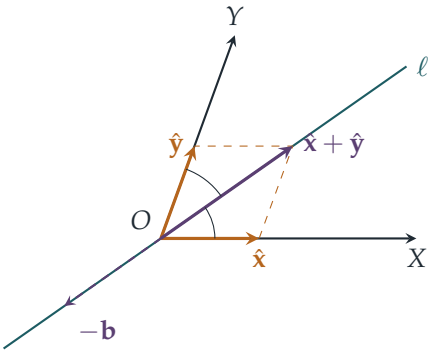
$$\mathbf{b} \cdot \hat{\mathbf{x}} = |\mathbf{x}||\mathbf{y}|\,(1 + \cos\theta) = \mathbf{b} \cdot \hat{\mathbf{y}},$$

so the angle $\alpha$ between $\mathbf{b}$ and $\overrightarrow{OX}$ and the angle $\beta$ between $\mathbf{b}$ and $\overrightarrow{OY}$ satisfy

$$\cos\alpha = \frac{\mathbf{b} \cdot \hat{\mathbf{x}}}{|\mathbf{b}|} = \frac{|\mathbf{x}||\mathbf{y}|(1 + \cos\theta)}{|\mathbf{b}|} = \frac{\mathbf{b} \cdot \hat{\mathbf{y}}}{|\mathbf{b}|} = \cos\beta.$$

Both angles lie in $[0, \pi]$, where cosine is one-to-one, so $\alpha = \beta$. Also $1 + \cos\theta > 0$, so $\cos\alpha > 0$ and both angles are less than $90°$. Hence $\mathbf{b}$ is a bisecting vector for angle $XOY$.

**Other bisecting vectors.**




Since $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ have equal length, the parallelogram they span is a rhombus, and its diagonal $\hat{\mathbf{x}} + \hat{\mathbf{y}}$ splits angle $XOY$ into two equal halves. In the plane $XOY$, the only directions making equal angles with $OX$ and $OY$ are the two directions along this diagonal line $\ell$. If we rotate a vector off $\ell$ it moves closer to one arm and further from the other, so the angles become unequal. Along $\ell$, the direction of $\mathbf{b}$ makes angle $\frac{1}{2}\theta < 90°$ with each arm. The opposite direction makes angle $180° - \frac{1}{2}\theta > 90°$ with each, so it is ruled out. So any bisecting vector points along $\ell$ in the same direction as $\mathbf{b}$, which means it is a positive multiple of $\mathbf{b}$.

The same argument in algebra: any vector in the plane is $\mathbf{v} = p\hat{\mathbf{x}} + q\hat{\mathbf{y}}$. Equal angles means $\mathbf{v} \cdot \hat{\mathbf{x}} = \mathbf{v} \cdot \hat{\mathbf{y}}$, and

$$\mathbf{v} \cdot \hat{\mathbf{x}} - \mathbf{v} \cdot \hat{\mathbf{y}} = (p + q\cos\theta) - (p\cos\theta + q) = (p - q)(1 - \cos\theta).$$

Since $\cos\theta \neq 1$, this forces $p = q$, so $\mathbf{v} = p(\hat{\mathbf{x}} + \hat{\mathbf{y}})$. The angles are acute only if $p > 0$.

#### Part (i)(b)

The point $B$ lies on the line $XY$ exactly when $\overrightarrow{OB} = (1 - t)\mathbf{x} + t\mathbf{y}$ for some real $t$. We need

$$\lambda|\mathbf{y}|\,\mathbf{x} + \lambda|\mathbf{x}|\,\mathbf{y} = (1 - t)\mathbf{x} + t\mathbf{y}.$$

Since $\mathbf{x}$ and $\mathbf{y}$ are not parallel, we may compare coefficients: $\lambda|\mathbf{y}| = 1 - t$ and $\lambda|\mathbf{x}| = t$. Adding these gives

$$\lambda = \frac{1}{|\mathbf{x}| + |\mathbf{y}|}, \qquad t = \frac{|\mathbf{x}|}{|\mathbf{x}| + |\mathbf{y}|}.$$

Then $\overrightarrow{XB} = t(\mathbf{y} - \mathbf{x})$ and $\overrightarrow{BY} = (1 - t)(\mathbf{y} - \mathbf{x})$ with $0 < t < 1$, so $B$ lies between $X$ and $Y$ and

$$XB : BY = t : (1 - t) = |\mathbf{x}| : |\mathbf{y}| = OX : OY.$$

This is the angle bisector theorem: the bisector from $O$ divides $XY$ in the ratio of the adjacent sides.

**Part (i)(c)**

Here $\overrightarrow{OB}$ is a non-zero multiple of $\mathbf{b}$, so $OB \perp XY$ means $\mathbf{b} \cdot (\mathbf{y} - \mathbf{x}) = 0$. Using $\mathbf{b} = |\mathbf{x}||\mathbf{y}|(\hat{\mathbf{x}} + \hat{\mathbf{y}})$ and $\mathbf{y} - \mathbf{x} = |\mathbf{y}|\hat{\mathbf{y}} - |\mathbf{x}|\hat{\mathbf{x}}$,

$$\begin{aligned}\mathbf{b} \cdot (\mathbf{y} - \mathbf{x}) &= |\mathbf{x}||\mathbf{y}|(|\mathbf{y}|\cos\theta + |\mathbf{y}| - |\mathbf{x}| - |\mathbf{x}|\cos\theta) \\ &= |\mathbf{x}||\mathbf{y}|\,(|\mathbf{y}| - |\mathbf{x}|)(1 + \cos\theta).\end{aligned}$$

Now $|\mathbf{x}|, |\mathbf{y}| > 0$ and $1 + \cos\theta > 0$, so this is zero only if $|\mathbf{y}| = |\mathbf{x}|$. That is, $OX = OY$, and triangle $XOY$ is isosceles.

**Part (ii)**

Let $\mathbf{p}$, $\mathbf{q}$, $\mathbf{r}$ be the unit vectors along $\overrightarrow{OP}$, $\overrightarrow{OQ}$, $\overrightarrow{OR}$. No three of the points are collinear, so in particular $O, P, Q$ are not, and part (i)(a) applies to each angle. The chosen bisecting vectors must therefore be

$$\mathbf{u} = \alpha(\mathbf{p} + \mathbf{q}), \qquad \mathbf{v} = \beta(\mathbf{q} + \mathbf{r}), \qquad \mathbf{w} = \gamma(\mathbf{r} + \mathbf{p}),$$

for some $\alpha, \beta, \gamma > 0$, and all three are non-zero.

> **Ask yourself** The three bisecting vectors are arbitrary positive multiples. How can we say anything about all three angles at once?

The scale factors are positive, so they cannot change the sign of a dot product. Only the signs matter, since an angle between non-zero vectors is acute, right or obtuse according as their dot product is positive, zero or negative. Write $S = \mathbf{p} \cdot \mathbf{q} + \mathbf{q} \cdot \mathbf{r} + \mathbf{r} \cdot \mathbf{p}$. Using $\mathbf{q} \cdot \mathbf{q} = 1$,

$$\mathbf{u} \cdot \mathbf{v} = \alpha\beta\,(\mathbf{p} \cdot \mathbf{q} + \mathbf{p} \cdot \mathbf{r} + \mathbf{q} \cdot \mathbf{q} + \mathbf{q} \cdot \mathbf{r}) = \alpha\beta\,(1 + S).$$

In exactly the same way, using $\mathbf{r} \cdot \mathbf{r} = 1$ and $\mathbf{p} \cdot \mathbf{p} = 1$,

$$\mathbf{v} \cdot \mathbf{w} = \beta\gamma\,(1 + S), \qquad \mathbf{w} \cdot \mathbf{u} = \gamma\alpha\,(1 + S).$$

Since $\alpha\beta$, $\beta\gamma$, $\gamma\alpha$ are all positive, the three dot products all have the sign of $1 + S$. So the three angles are all acute if $1 + S > 0$, all right angles if $1 + S = 0$, and all obtuse if $1 + S < 0$. ■

## Checks

- Take $\mathbf{x} = (3, 0, 0)$ and $\mathbf{y} = (0, 4, 0)$. Then $\mathbf{b} = (12, 12, 0)$, which bisects the right angle. With $\lambda = \frac{1}{7}$ we get $B = (\frac{12}{7}, \frac{12}{7}, 0)$, which lies on $\frac{x}{3} + \frac{y}{4} = 1$, and $XB : BY = 3 : 4$.
- All three cases of part (ii) really occur. If $P$, $Q$, $R$ are spread at $120^\circ$ to each other in a plane, then $S = -\frac{3}{2}$ and the bisectors meet at $120^\circ$ (obtuse). If $\mathbf{p}, \mathbf{q}, \mathbf{r}$ point to three vertices of a regular tetrahedron centred at $O$, each dot product is $-\frac{1}{3}$, so $S = -1$ and all three angles are right angles. Three nearly parallel vectors give acute angles.
- The results of both parts were also confirmed by computer on many thousands of random configurations.

## Common slips

- Forgetting that $\cos\theta \neq \pm 1$ has to be used. The factor $1+\cos\theta$ in (i)(c) and the factor $1-\cos\theta$ in the uniqueness argument are non-zero only because the points are not collinear.
- In part (ii), treating the bisecting vectors as specific vectors such as $|\mathbf{x}|\mathbf{y}+|\mathbf{y}|\mathbf{x}$. The question lets them be chosen freely, so you have to quote part (i)(a) to say they are positive multiples of these.

## Beyond STEP

Writing a vector as $|\mathbf{x}|\hat{\mathbf{x}}$ and working with unit vectors is a habit that pays off throughout linear algebra. The identity in (ii) is a small piece of the geometry of the *Gram matrix* of $\mathbf{p}, \mathbf{q}, \mathbf{r}$, whose entries are their pairwise dot products. The fact that $|\mathbf{p}+\mathbf{q}+\mathbf{r}|^2 = 3+2S \geq 0$ shows that $S \geq -\frac{3}{2}$, so the obtuse case has a limit. Strang's *Introduction to Linear Algebra* develops dot products, projections and Gram matrices in this spirit.

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